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Final Project Report

Decoding Boston Safety

An In-Depth Exploration of Crime Trends and Safety insights in Boston

Generated by ChatGPT

ABSTRACT

Boston is one of the largest cities in the US and also known as an academic city with many renowned universities. People keep coming in, and the city is constantly changing. In such a city, one of the main interests/concerns for both newcomers and long-time residents would be how safe Boston is–because it directly affects the quality of life and becomes the criteria for finding a place for housing. The goal of this data visualization is to analyze Boston's safety from a data perspective.

Updated May 14, 2024

Description

I moved to Boston a few years ago. I had heard that Boston is a safe city because of its youthful population and its presence as a college town, and indeed, I have not experienced any physical danger so far. However, some of my friends have encountered or been the targets of criminal incidents, and I feel there is a discrepancy between the reality and our perception.

The factor "safety" is even more important for newcomers to Boston. Just knowing where is safe (or where is dangerous) can make the experience in Boston much better.

Therefore, this project namely examines Boston's public safety. We often hear that downtown Boston, Roxbury, and Dorchester are dangerous, etc., but how much of this is true? What areas are truly dangerous and at what times of the day? What kind of crimes are happening? Based on the crime incident reports provided by the City of Boston, this project analyzes when, where, and what kind of crime cases are occuring, and seeks trends and measures.

The overall structure is as follows:

1. **Comparing Boston to other cities**
   Visualization of how Boston is positioned in terms of public safety compared to other large cities and other Massachusetts cities.
2. **The number of crimes in Boston**
   This section tells you how much public safety has changed over the past 5 years. It also shows if each crime type has followed the same trend or not.
3. **When crime is likely to occur**
   Once you know when crime is most likely to occur, it becomes much easier to take action. In this section, you see the plots showing the number of crimes by month, week, and day time.
4. **Areas where crime is likely to occur**

It is also important to know the areas where crime is more likely to occur. The crime incident reports have information on which crimes were handled by which police jurisdiction. I mapped out these jurisdictions (or districts), and then color-coded each jurisdiction to show how many crime cases happened in each area.

Datasets

I first had to start with the data processing of the crime incident reports by the City of Boston. Since the data is stored separately by year in the database and its format is also different in some places, I standardized it beforehand.

Additionally, organizing the data was another important process in this project. The report is a collection of a number of cases handled by the police, not all of which directly involve us. For example, some cases include towing a car or assisting children. Since I want this project to focus on crimes that could happen to us, I excluded those that are not considered "common" crimes based on my own judgment.

Crime rates data:

Boston	https://www.neighborhoodscout.com/ma/boston/crime
DC	https://www.neighborhoodscout.com/dc/washington/crime
Chicago	https://www.neighborhoodscout.com/il/chicago/crime
Seattle	https://www.neighborhoodscout.com/wa/seattle/crime
New York	https://www.neighborhoodscout.com/ny/new-york/crime
San Francisco	https://www.neighborhoodscout.com/ca/san-francisco/crime
Austin	https://www.neighborhoodscout.com/tx/austin/crime
Philadelphia	https://www.neighborhoodscout.com/pa/philadelphia/crime
Phoenix	https://www.neighborhoodscout.com/az/phoenix/crime
Los Angeles	https://www.neighborhoodscout.com/ca/los-angeles/crime

Amherst	https://www.neighborhoodscout.com/ma/amherst/crime
Lowell	https://www.neighborhoodscout.com/ma/lowell/crime
Dartmouth	https://www.neighborhoodscout.com/ma/dartmouth/crime
Springfield	https://www.neighborhoodscout.com/ma/springfield/crime
Worcester	https://www.neighborhoodscout.com/ma/worcester/crime
Lynn	https://www.neighborhoodscout.com/ma/lynn/crime
Framingham	https://www.neighborhoodscout.com/ma/framingham/crime
Plymouth	https://www.neighborhoodscout.com/ma/plymouth/crime

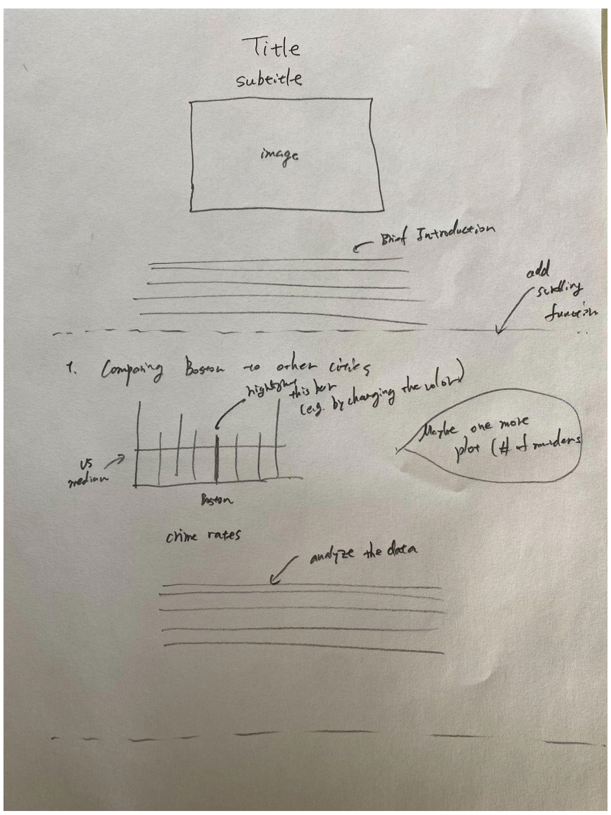
Cambridge https://www.neighborhoodscout.com/ma/cambridge

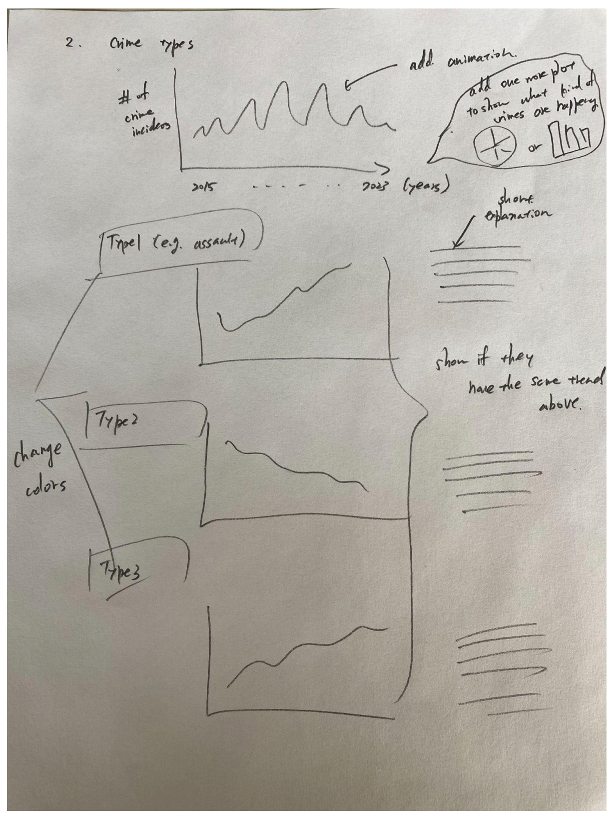
Crime types and date data:
https://data.boston.gov/dataset/crime-incident-reports-august-2015-to-date-source-new-system

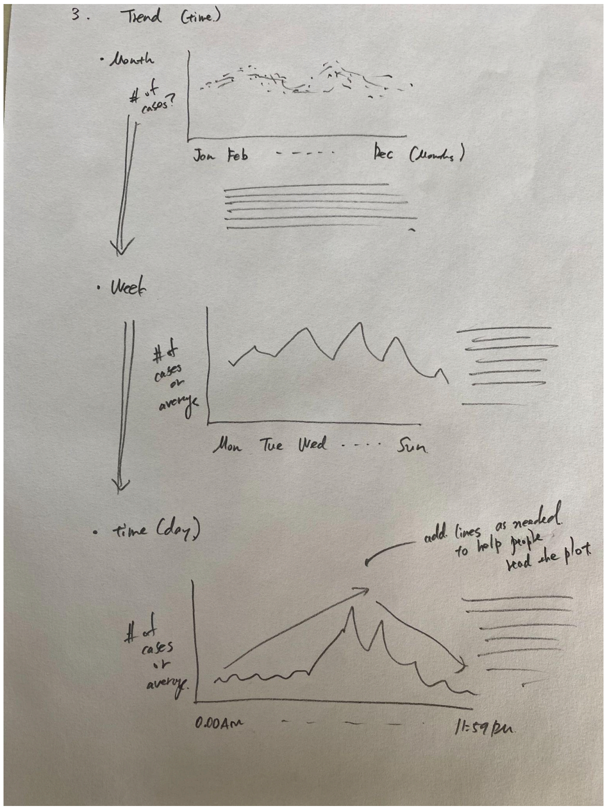
Crime area data:
https://bostonopendata-boston.opendata.arcgis.com/datasets/9a3a8c427add450eaf45a470245680fc/explore
https://police.boston.gov/districts/

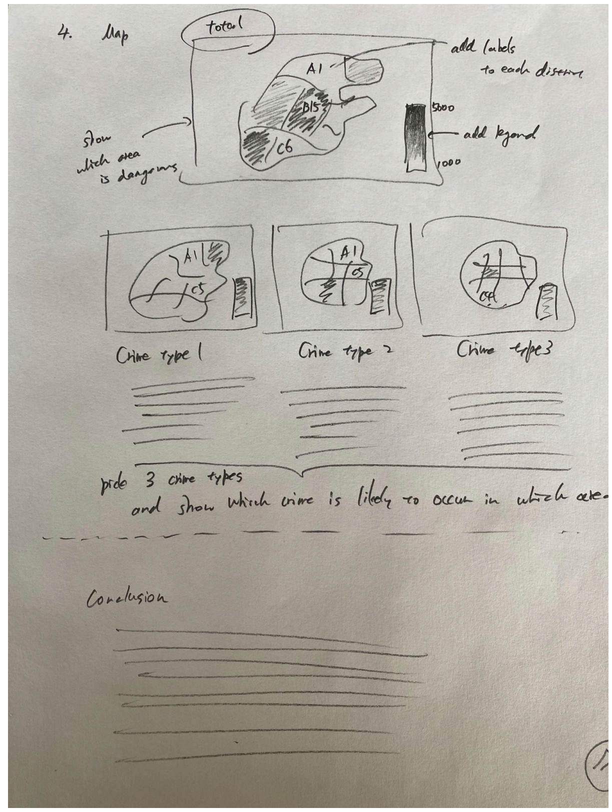
Boston Statistics:
https://data.census.gov/profile/Boston_city,_Massachusetts?g=160XX00US2507000

Paper Sketches









Technologies, Development and Deployment

All plots in this project were created with d3.js. I tried animating all of them as much as possible using the code in the D3 Gallery and with the help of ChatGPT.

For the animations, I made use of Observer API so that the animation for each plot would only start when we reach each plot. Every plot has a unique id, and Observer API uses that to determine when to start the animation.

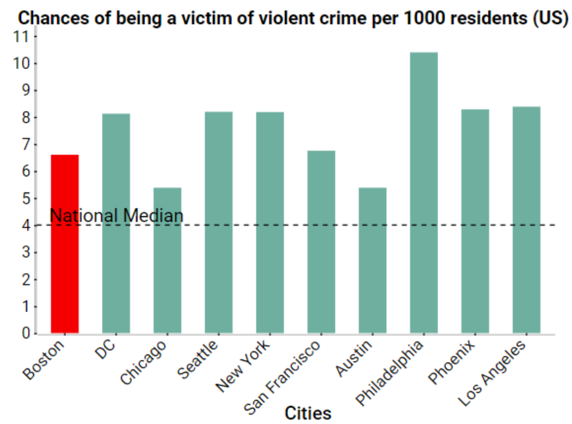
As for the maps, I used a geojson file provided by the City of Boston. This works very well with d3.js, and the data was retrieved as easily as reading a csv file.

Finally, regarding the code structure, the final project repository contains three folders: *data*, *data_processing*, *js*, and the main file: *index.html*. *data_processing* contains ipynb files that create csv files for the plots, and *data* stores the original data and the csv files created. *js*, as the name suggests, is where all javascript files for the plots are stored. Last but not least, *index.html* is hosted by Github as usual.

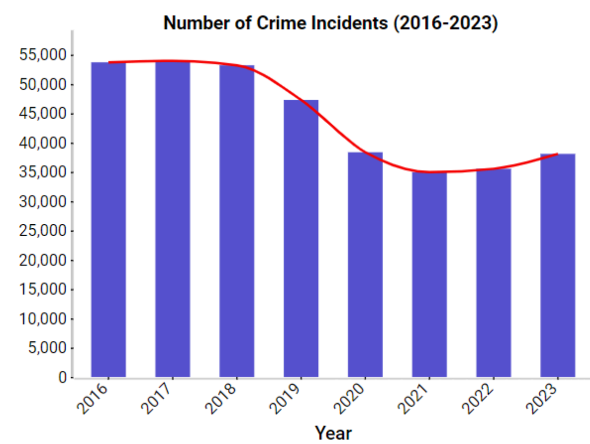
Final Product

The first thing that appears on the first page is the intro section. I put the title and an image here so that readers can tell at a glance what this project is all about. I also included the sound of a police siren to attract readers' attention from an auditory aspect as well.

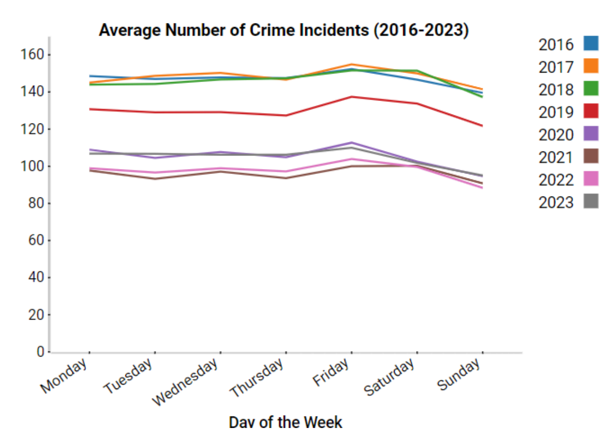
After a brief introduction, what you see is comparisons between Boston and other cities. Since I wanted to show the chances of being a victim of crime for each city, I selected a bar chart. As a guide to show what the number looks like, I have shown the national and Massachusetts median for each plot. Additionally, in order to emphasize Boston's position in the plots, Boston's bar is shown in red.



The next section is about the change in the number of crimes over the years. Since we are particularly interested in the increase/decrease trend, the plot draws a line connecting each bar. This makes it easier to read the changes in the number of crimes than to simply follow the height of each bar by eye. Furthermore, in order to entertain readers, the bars and lines are displayed from the left with sound effects. In terms of color, I have added some color adjustment, such as using green for a crime type that has been declining.

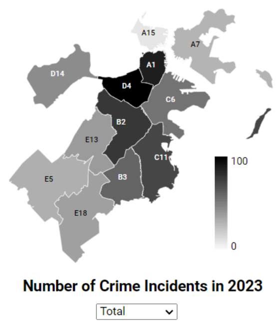


Next comes the section on crime trends. The first plot shows the monthly trend. It is a scatter plot where time and the number of crimes occurred on each day are shown on the horizontal axis and the vertical axis respectively. Since I wanted to show multiple years of data, the plot assigns each year a unique color and shows its mapping on the right as a legend.



The weekly trend comes after the monthly trend. Here, the horizontal axis can only take 7 values, therefore the plot shows the average number of cases occurred on each day of the week. Again, the multi-year data is represented with multiple colors. The same was done for the hourly trend.

Finally, this article ends with a map showing the relationships between the number of crimes and the areas. I once considered visualizing the number of crimes by adding bars or circles on the map, but for the sake of readability, I kept the map in 2D and went in the direction using color contrast. Also, to keep the map as concise as possible, I decided to leave only the district code on the map and show the name of each place in a table. Additionally, I created a drop-down box to allow people to choose crime types as well.



Code	Location
A1	Downtown
A15	Charlestown
A7	East Boston
B2	Roxbury
B3	Mattapan
C6	South Boston
C11	Dorchester
D4	South End
D14	Brighton
E5	West Roxbury
E13	Jamaica Plain
E18	Hyde Park

For the overall design, I wanted to make this project in a newspaper article style, and I referenced online articles from the Boston Globe and the New York Times.

<PROJECT LINK>
https://kenichi-maeda.github.io/cs617/final_project/
<PROJECT CODE>
https://github.com/kenichi-maeda/cs617/tree/main/final_project

References

<Code>
https://tomordonez.com/d3-bar-chart-title-and-labels/
https://developer.mozilla.org/en-US/docs/Web/API/Intersection_Observer_API
https://gist.github.com/almccon/a53831a573911d0a875821c5f9fac6be
https://observablehq.com/@onoratod/animate-a-path-in-d3
https://stackoverflow.com/questions/52545287/d3-draw-line-using-stroke-dasharray-and-stroke-dashoffset
https://stackoverflow.com/questions/55397871/how-to-fix-join-is-not-a-function-when-trying-join-after-selectall
https://stackoverflow.com/questions/22735424/how-to-animate-a-path-along-with-its-points-in-d3

https://d3-graph-gallery.com/index.html
https://d3-graph-gallery.com/graph/scatter_basic.html
https://d3-graph-gallery.com/graph/barplot_basic.html
https://d3-graph-gallery.com/graph/line_basic.html
https://d3-graph-gallery.com/graph/backgroundmap_basic.html
https://mbtaviz.github.io/

<Sound>
https://pixabay.com/sound-effects/search/click/?pagi=3
https://mixkit.co/free-sound-effects/police/

<Debugging/Ideas/Image>
ChatGPT